Unit C: Two-Sample Inference

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INTRODUCTION

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Example 1

In families with exactly two daughters, who is the taller sister (on average) when they reach adulthood, the older sister or the younger sister? Or are they same height (on average)?

Randomly select pairs of sisters.

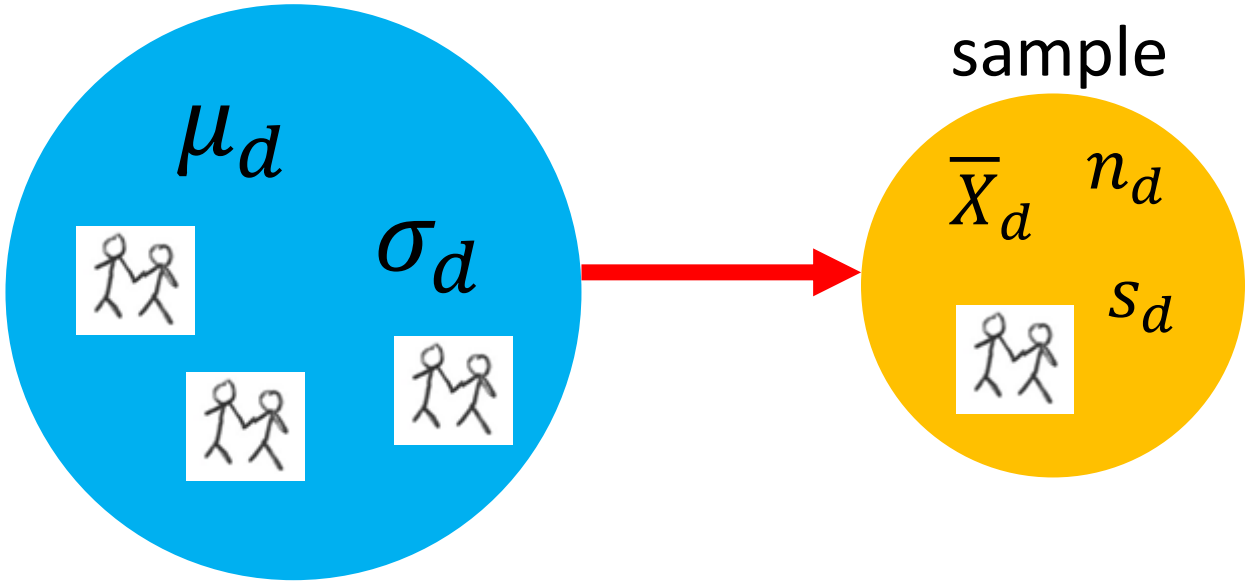
x = height of older sister on her 25th birthday

y = height of younger sister on her 25th birthday

$d = x - y$

hypothesis test for μ_d (paired differences)

a population of pairs



$$H_0: \mu_d = 0$$

$$H_a: \mu_d \neq 0$$

$\overline{X}_d$ estimates μ_d

Example 2

Do mothers who smoke during pregnancy give birth to lighter babies?

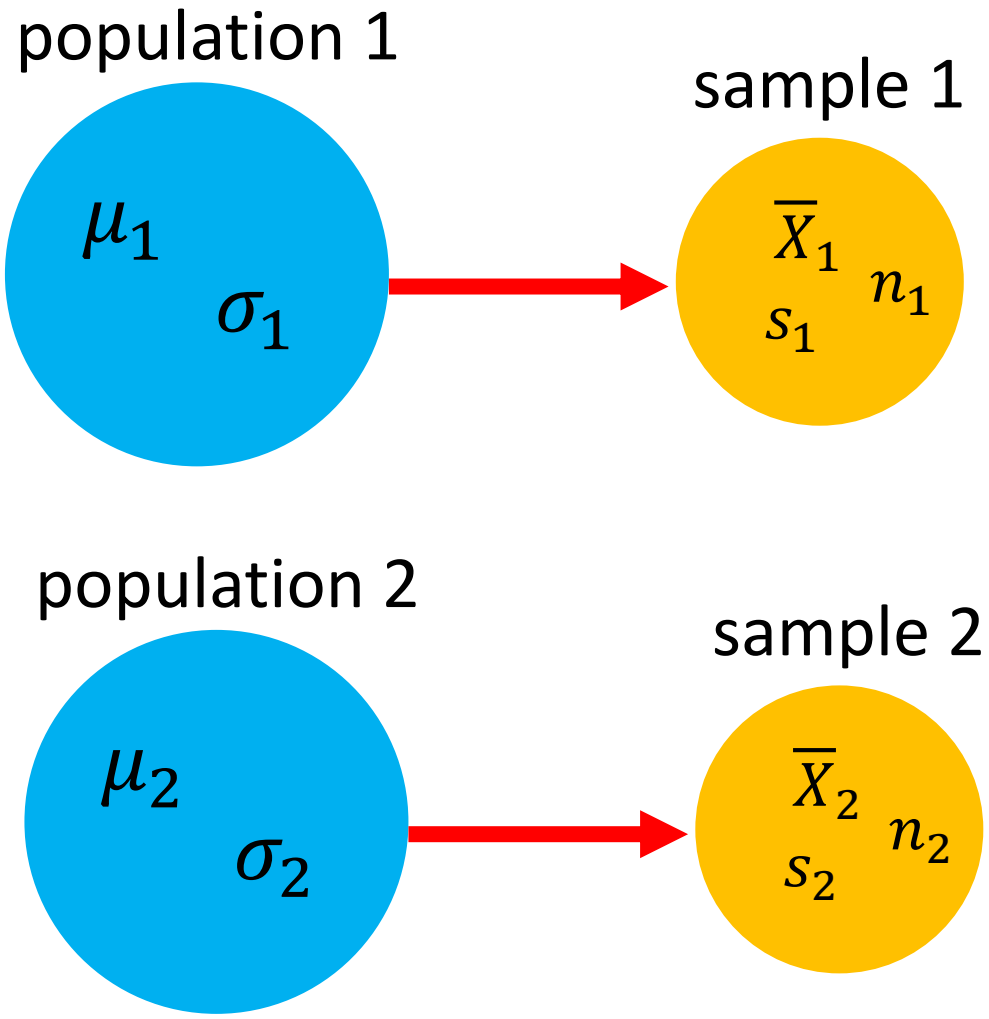
μ_1 = population mean birthweight among babies born to smoking mothers

μ_2 = population mean birthweight among babies born to non-smoking mothers

$$H_0: \mu_1 = \mu_2 \rightarrow \mu_1 - \mu_2 = 0$$

$$H_a: \mu_1 < \mu_2 \rightarrow \mu_1 - \mu_2 < 0$$

two-sample hypothesis test for $\mu_1 - \mu_2$ (independent samples)



$\overline{X}_1 - \overline{X}_2$

estimates

$\mu_1 - \mu_2$

Example 3

Do people in their 70's vote more often than people in their 20's?

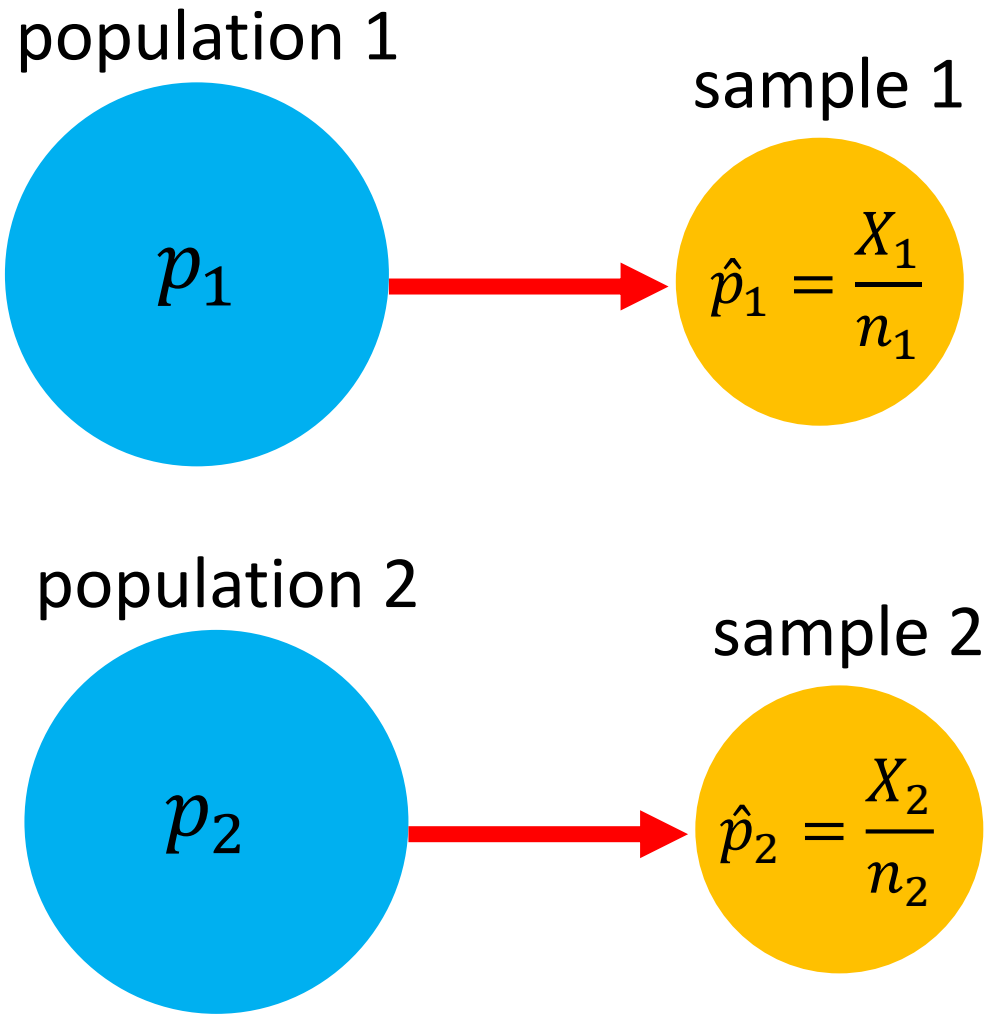
p_1 = population proportion among people aged 70-80 who voted in the last election

p_2 = population proportion among people aged 20-30 who voted in the last election

$$H_0: p_1 = p_2 \rightarrow p_1 - p_2 = 0$$

$$H_a: p_1 > p_2 \rightarrow p_1 - p_2 > 0$$

two-sample hypothesis test for $p_1 - p_2$ (independent samples)



$\hat{p}_1 - \hat{p}_2$

estimates

$p_1 - p_2$

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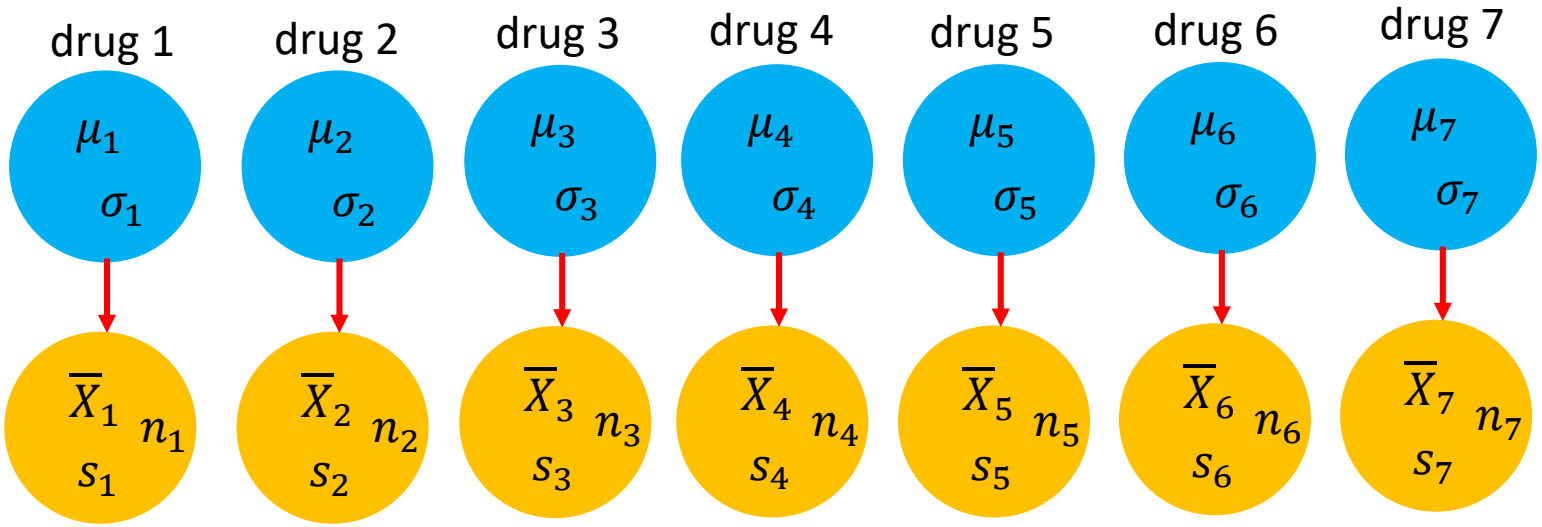
Example 4

Drug researchers are considering 7 different blood pressure medications and want to know which one is most effective.

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6 = \mu_7$$

$$H_a: H_0 \text{ is false}$$

One-way ANOVA (Analysis of Variance)



$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6 = \mu_7$$

$$H_a: H_0 \text{ is false}$$

$n =$ overall sample size $= n_1 + n_2 + \cdots + n_k$

$\overline{X} =$ overall sample mean among all k samples

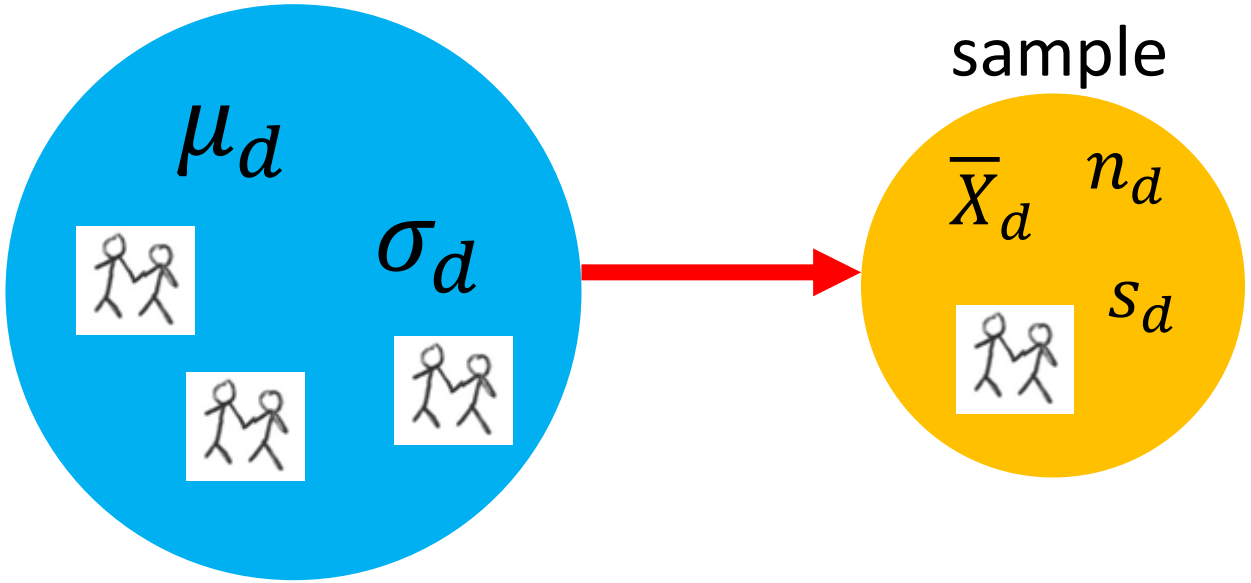
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PAIRED DIFFERENCES T TEST

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hypothesis test for μ_d (paired differences)

a population of pairs



$$H_0: \mu_d = 0$$

$\overline{X}_d$ estimates μ_d

Test statistic recipe: $T = \frac{(\text{estimator}) - (\text{hypothesized parameter value})}{SD(\text{estimator})}$

$$T = \frac{\overline{X}_d - \mu_d}{s_d / \sqrt{n_d}}$$

degrees of freedom = $n_d - 1$

A paired data hypothesis test for μ_d is very similar to a regular one-sample T-test for μ. Now the data set is a set of differences.

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example:

Does caffeine increase resting heart rate (RHR)? Conduct a paired T-test at α =.05. 15 volunteers were randomly selected to be part of a study. During two sessions on different days, test subjects were told to relax for 30 minutes after taking either a caffeine pill or a placebo pill. Their RHR was recorded 30 minutes after consuming the pill. Each test subject had a difference between the two heart rates:

d = (RHR while caffeinated) – (RHR after taking placebo)

The data set $\{ d_1, d_2, \ldots d_{15}\}$ has a sample mean of $\overline{X}_d = 8.1$ and a sample standard deviation of $s_d = 10.39$

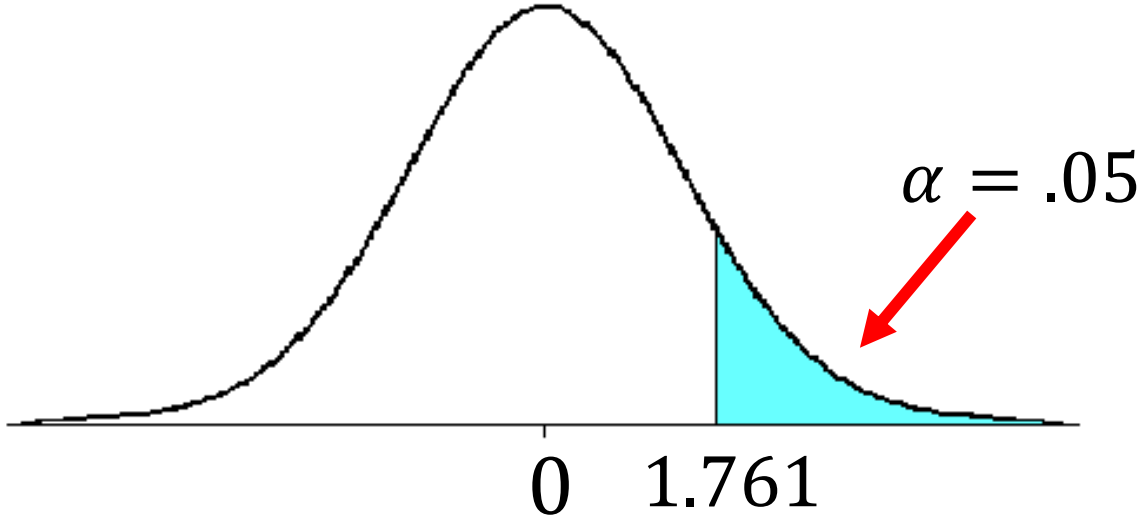
$$H_0\colon \mu_d = 0$$
$$H_a\colon \mu_d > 0$$

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$$H_0: \mu_d = 0$$
$$H_a: \mu_d > 0$$

degrees of freedom = $n_d - 1 = 15 - 1 = 14$

Rejection Region is right-tailed on a t-curve with 14 d.o.f.



d = (RHR while caffeinated) – (RHR after taking placebo)

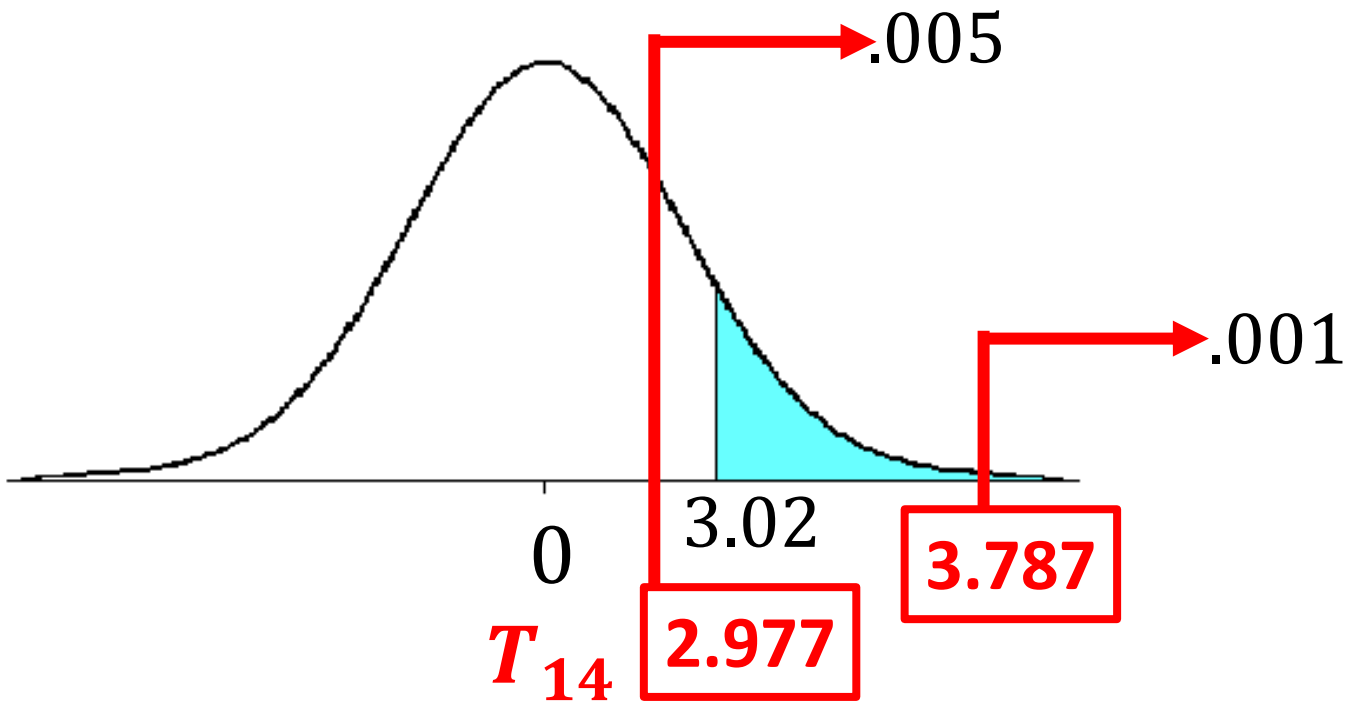
The data set $\{ d_1, d_2, \ldots d_{15}\}$ has a sample mean of $\overline{X}_d = 8.1$ and a sample standard deviation of $s_d = 10.39$

Test statistic:

$$T = \frac{\overline{X}_d - \mu_d}{s_d/\sqrt{n_d}} = \frac{8.1 - 0}{10.39/\sqrt{15}} \approx 3.02$$

Conclusion:

reject H_0
We have significant evidence that caffeine increases RHR.



Assumptions/requirements:

$\overline{X}_d$ normal (because population of differences is normal) or $\overline{X}_d$ approx. normal (because $n_d \geq 30$)

.001 < p-value < .005

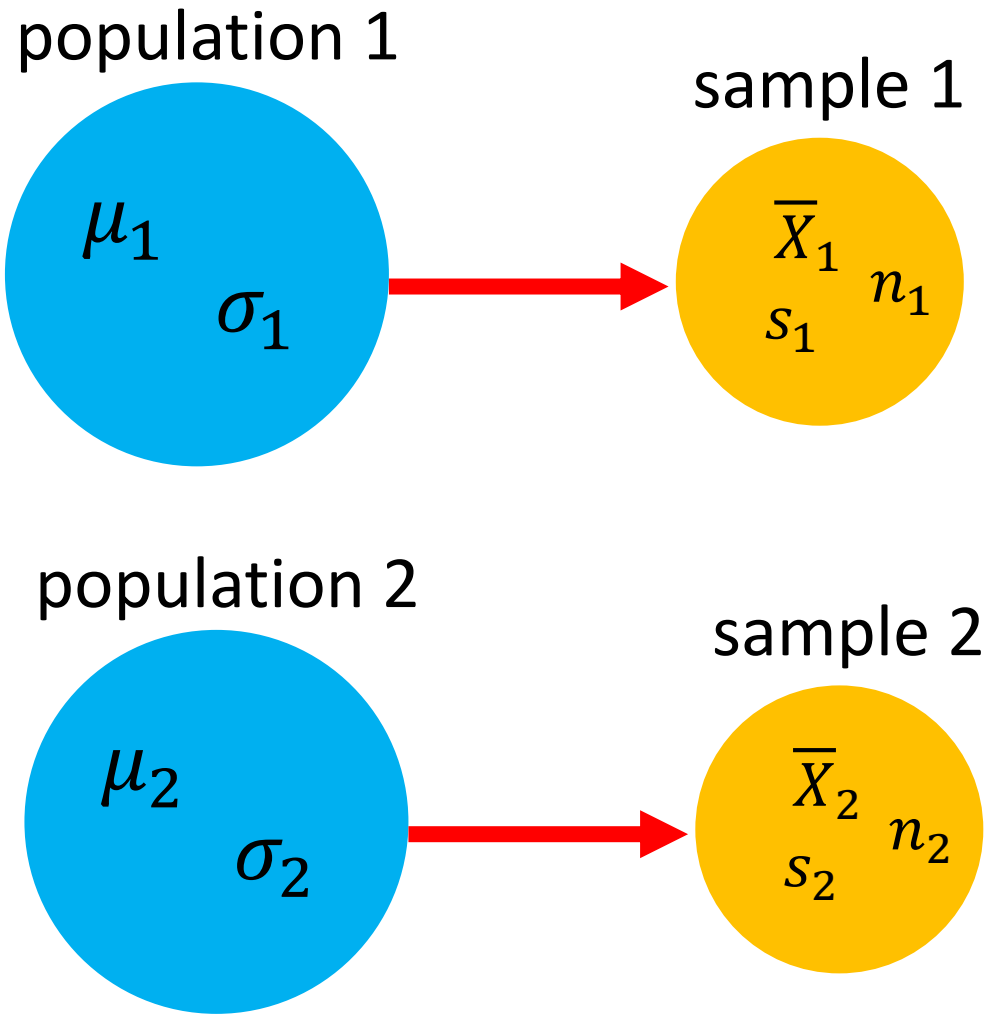
Since the sample size was 15 in this story, we must assume normality and examine a QQ Plot.

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TWO-SAMPLE T TEST FOR MEANS

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two-sample hypothesis test for $\mu_1 - \mu_2$ (independent samples)



$\overline{X}_1 - \overline{X}_2$

estimates

$\mu_1 - \mu_2$

$\overline{X}_1 - \overline{X}_2$ estimates $\mu_1 - \mu_2$

$\overline{X}_1$ and $\overline{X}_2$ are normal (because populations are normal) or approx. normal (because $n_1 \geq 30$ and $n_2 \geq 30$)

Therefore $\overline{X}_1 - \overline{X}_2$ is normal or approx. normal

Recall: a sample mean is unbiased for the population mean: $E(\overline{X}) = \mu$

$$E(\overline{X}_1 - \overline{X}_2) = E(\overline{X}_1) - E(\overline{X}_2) = \mu_1 - \mu_2$$

Therefore $\overline{X}_1 - \overline{X}_2$ is unbiased for $\mu_1 - \mu_2$

$\overline{X}_1 - \overline{X}_2$ estimates $\mu_1 - \mu_2$

we will assume $\sigma_1^2 = \sigma_2^2 = \sigma_p^2$

$$Var(\overline{X}_1 - \overline{X}_2) = Var(\overline{X}_1) + Var(\overline{X}_2) - 2\mathrm{Cov}(\overline{X}_1, \overline{X}_2)$$
$$= \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{\sigma_p^2}{n_1} + \frac{\sigma_p^2}{n_2} = \sigma_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)$$

Therefore $\boldsymbol{SD(\overline{X}_1 - \overline{X}_2) = \sqrt{\sigma_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = \sigma_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

Test statistic recipe: $T = \frac{(\text{estimator}) - (\text{hypothesized parameter value})}{SD(\text{estimator})}$

$$T = \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

How do we estimate the common variance/standard deviation?

$$\sigma_1^2 = \sigma_2^2 = \sigma_p^2 \quad \rightarrow \quad \sigma_1 = \sigma_2 = \sigma_p$$

$$s_p \approx \sigma_p$$

How do we calculate the pooled sample standard deviation?

$$\boldsymbol{s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}}$$

$$s^2 = \frac{\sum(X - \overline{X})^2}{n - 1} \rightarrow (n - 1)s^2 = \sum(X - \overline{X})^2$$

$$(n_1 - 1) + (n_2 - 1) = \boldsymbol{n_1 + n_2 - 2 \text{ degrees of freedom}}$$

example:

μ_1 = population mean weight among adult golden retrievers
μ_2 = population mean weight among adult rottweilers

Weights (pounds) were recorded for two independent random samples of $n_1 = 8$ golden retrievers and $n_2 = 14$ rottweilers. Do we have significant evidence (at α =.05) that the mean weight for these two populations are unequal?

Sample 1 (golden retrievers)
$n_1 = 8$, $\overline{X}_1$ = 97.2, s_1 = 12.1

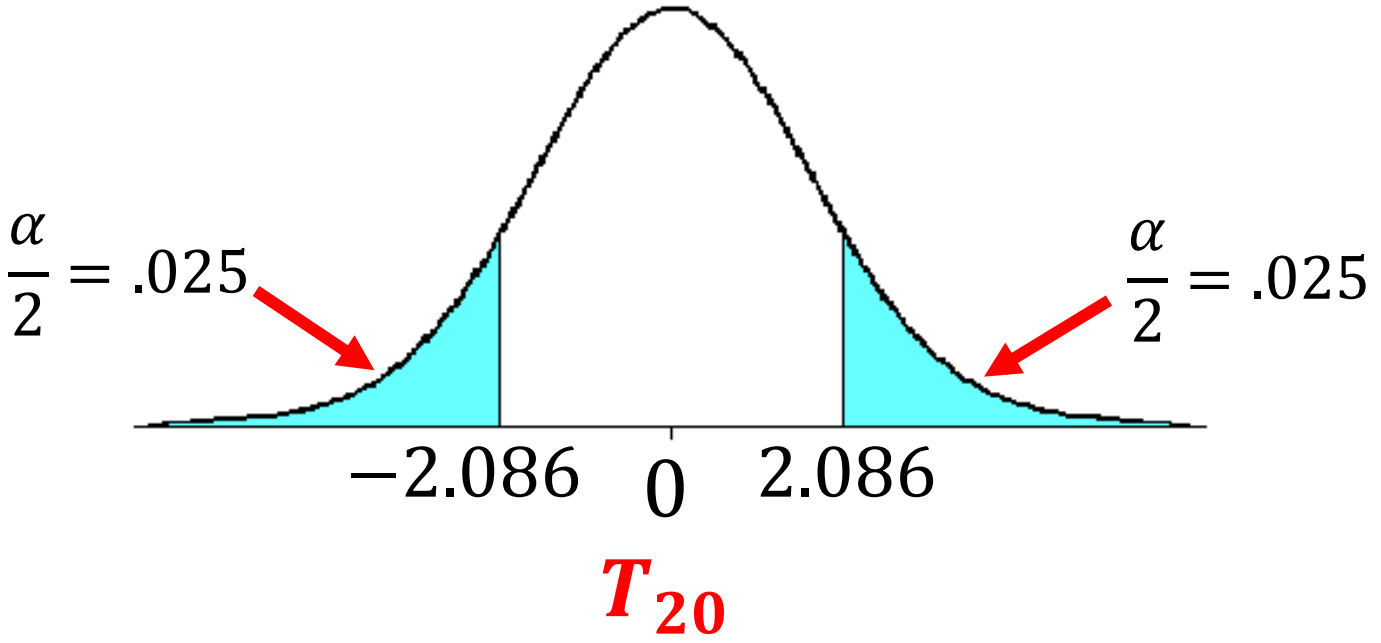
Sample 2 (rottweilers)
$n_2 = 14$, $\overline{X}_2$= 111.5, s_2 = 14.8

$$H_0: \mu_1 = \mu_2 \rightarrow \mu_1 - \mu_2 = 0$$
$$H_a: \mu_1 \neq \mu_2 \rightarrow \mu_1 - \mu_2 \neq 0$$

$$H_0\colon \mu_1 = \mu_2 \rightarrow \mu_1 - \mu_2 = 0$$
$$H_a\colon \mu_1 \neq \mu_2 \rightarrow \mu_1 - \mu_2 \neq 0$$

degrees of freedom = $n_1 + n_2 - 2 = 8 + 14 - 2 = 20$

Rejection Region is two-tailed on a t-curve with 20 d.o.f.



Sample 1 (golden retrievers)

$n_1 = 8$, $\overline{X}_1 = 97.2$, $s_1 = 12.1$

Sample 2 (rottweilers)

$n_2 = 14$, $\overline{X}_2 = 111.5$, $s_2 = 14.8$

Pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{7 * (12.1)^2 + 13 * (14.8)^2}{8 + 14 - 2}} \approx 13.91$$

Test statistic:

$$T = \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(97.2 - 111.5) - 0}{13.91\sqrt{\frac{1}{8} + \frac{1}{14}}} \approx -2.32$$

Conclusion:

reject H_0

We have significant evidence that adult golden retrievers and adult rottweilers have different weights, on average.

Assumptions/requirements:

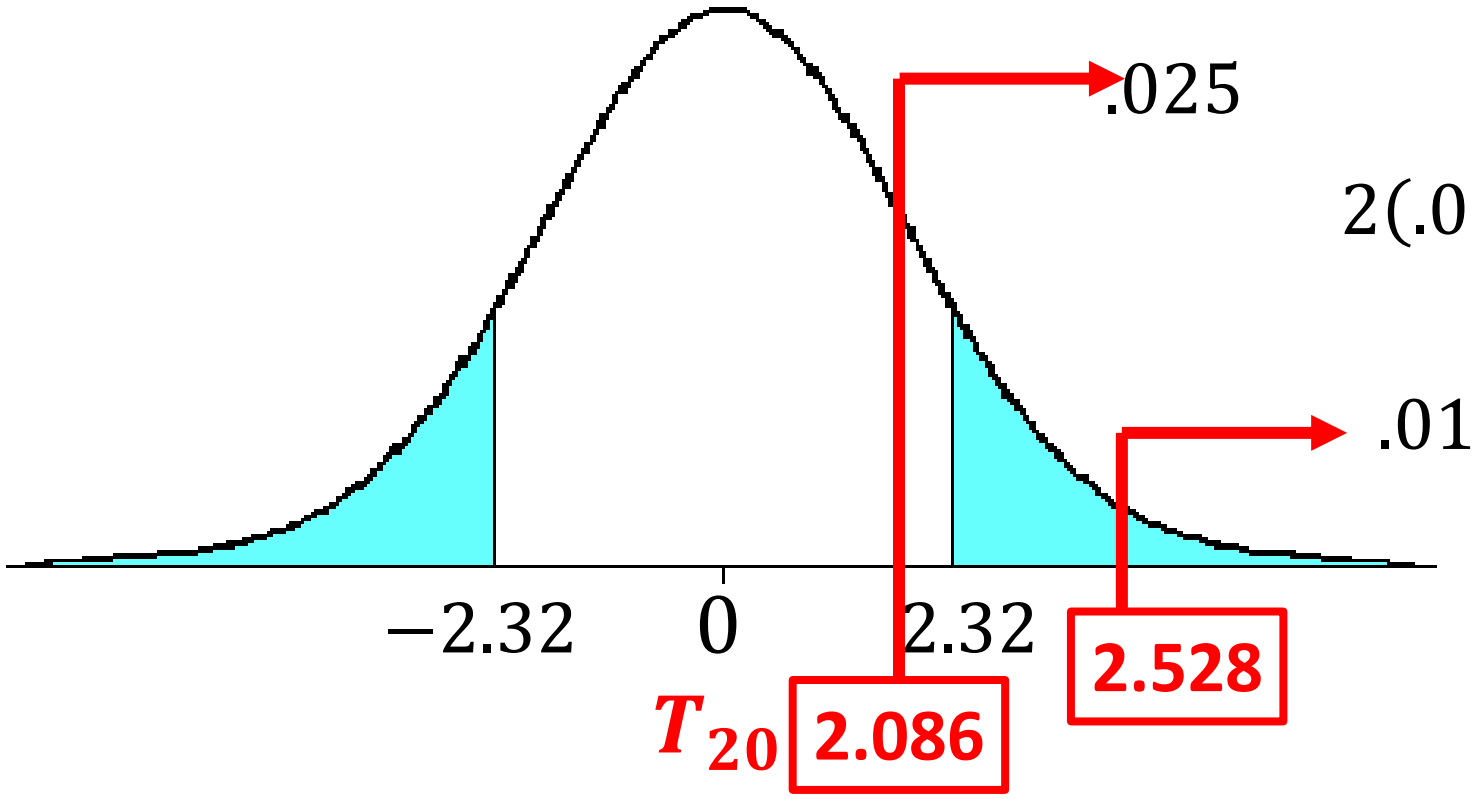
independent samples

$\sigma_1^2 = \sigma_2^2$

$\overline{X}_1$ and $\overline{X}_2$ normal (because populations are normal) or

$\overline{X}_1$ and $\overline{X}_2$ approx. normal (because $n_1 \geq 30$ and $n_2 \geq 30$)

Since the sample sizes in this story are small, we must assume normality for both populations and examine QQ Plots.



$$.01 < \frac{\text{p}-\text{value}}{2} < .025$$

$$2(.01) < 2\left(\frac{\text{p}-\text{value}}{2}\right) < 2(.025)$$

$$.02 < \text{p}-\text{value} < .05$$

Note:

Some courses/textbooks also cover the **Satterthwaite** option that does not require the equal variances assumption.

$$\sigma_1^2 \neq \sigma_2^2$$

independent samples

$\overline{X}_1$ and $\overline{X}_2$ normal (because populations are normal) or

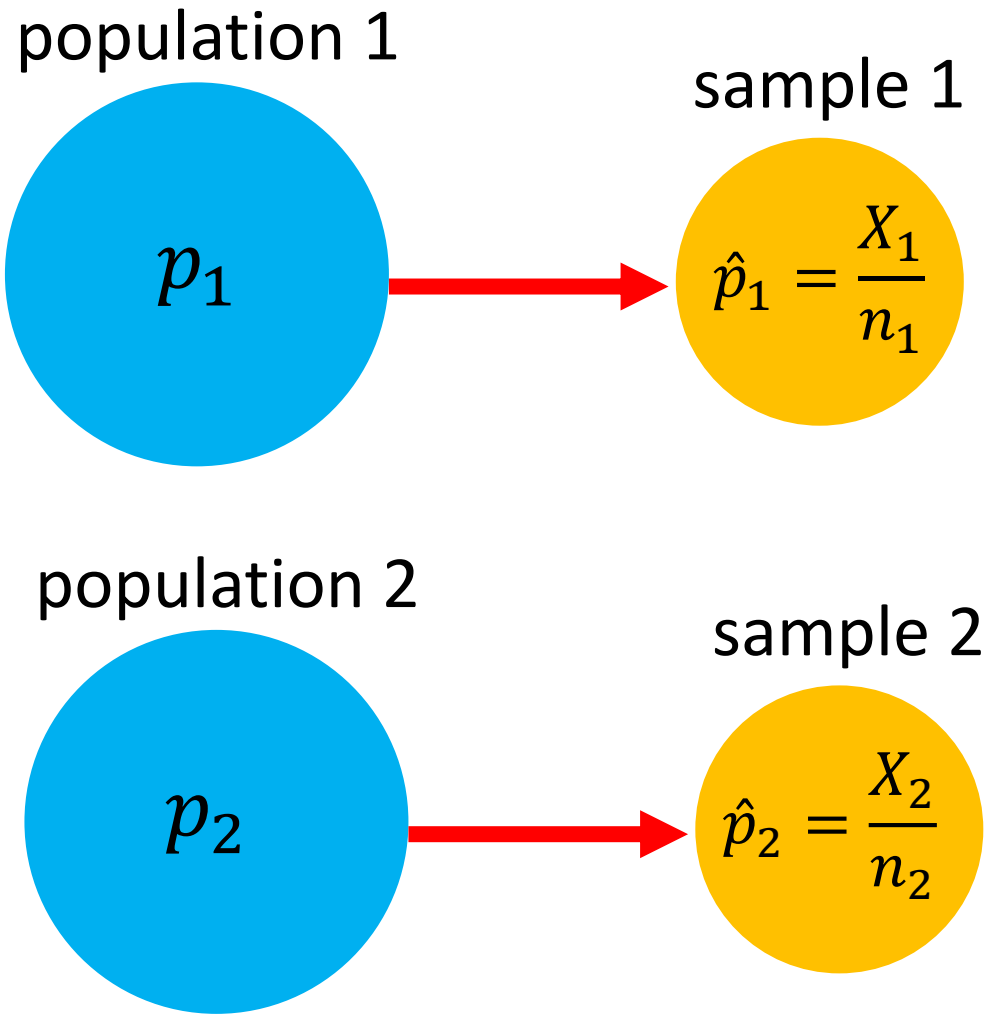
$\overline{X}_1$ and $\overline{X}_2$ approx. normal (because $n_1 \geq 30$ and $n_2 \geq 30$)

$$T = \frac{\left(\overline{X}_1 - \overline{X}_2\right) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

TWO-SAMPLE Z TEST FOR PROPORTIONS

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two-sample hypothesis test for $p_1 - p_2$ (independent samples)



$\hat{p}_1 - \hat{p}_2$

estimates

$p_1 - p_2$

H_0: $p_1 = p_2 \rightarrow p_1 - p_2 = 0$

$\hat{p}_1 - \hat{p}_2$ estimates $p_1 - p_2$

If $p_1 = p_2$ let's call the common value p

$$\hat{p} = \frac{X_1 + X_2}{n_1 + n_2}$$

Test statistic recipe: $Z = \frac{(\text{estimator}) - (\text{hypothesized parameter value})}{SD(\text{estimator})}$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}(1-\hat{p})}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Example

A new drug is developed that will hopefully help patients who suffer from migraine headaches. 200 patients are randomly selected and then randomly split into a treatment group and a placebo group. $\alpha = .05$

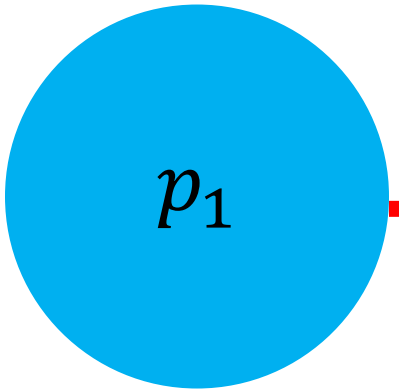
p_1 = population proportion among patients who will report an improvement after taking the drug

p_2 = population proportion among patients who will report an improvement after taking the placebo

$$H_0: p_1 = p_2 \rightarrow p_1 - p_2 = 0$$

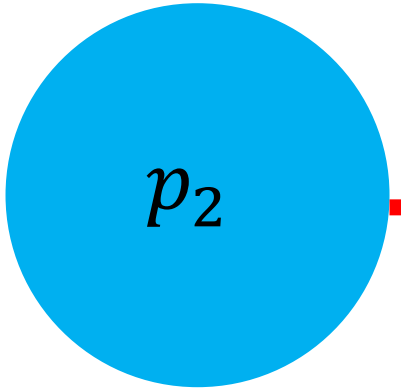
$$H_a: p_1 > p_2 \rightarrow p_1 - p_2 > 0$$

treatment



$$\hat{p}_1 = \frac{X_1}{n_1} = \frac{42}{100}$$

placebo



$$\hat{p}_2 = \frac{X_2}{n_2} = \frac{24}{100}$$

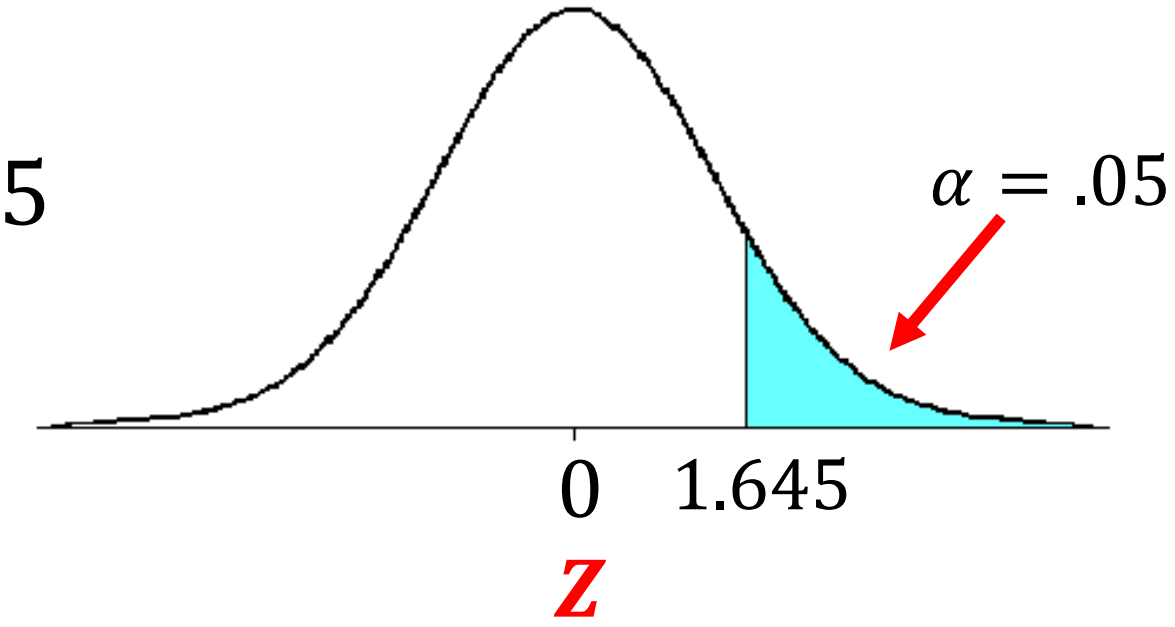
$$H_0: p_1 = p_2 \rightarrow p_1 - p_2 = 0$$

$$H_a: p_1 > p_2 \rightarrow p_1 - p_2 > 0$$

$$H_0: p_1 = p_2 \rightarrow p_1 - p_2 = 0$$
$$H_a: p_1 > p_2 \rightarrow p_1 - p_2 > 0$$

Rejection Region is right-tailed on the Z-curve.

RR: $Z > 1.645$



treatment

$$\hat{p}_1 = \frac{X_1}{n_1} = \frac{42}{100}$$

placebo

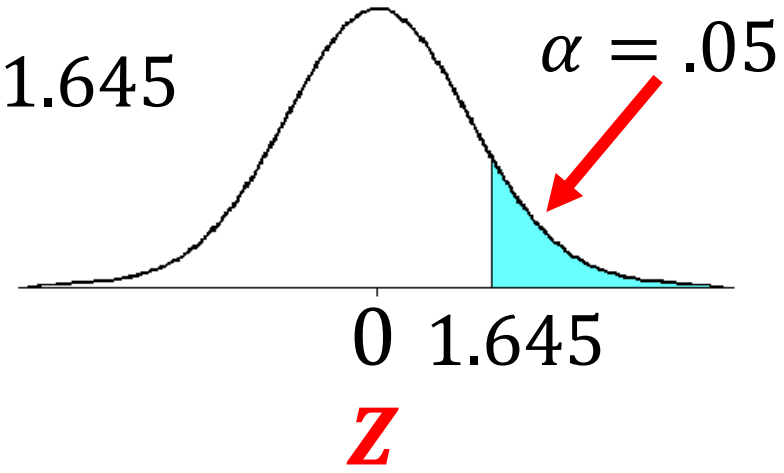
$$\hat{p}_2 = \frac{X_2}{n_2} = \frac{24}{100}$$

$$\hat{p} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{42 + 24}{100 + 100} = \frac{66}{200} = 0.33$$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}(1-\hat{p})}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(0.42 - 0.24) - 0}{\sqrt{0.33(0.67)}\sqrt{\frac{1}{100} + \frac{1}{100}}} \approx 2.71$$

$$H_0: p_1 = p_2 \rightarrow p_1 - p_2 = 0$$
$$H_a: p_1 > p_2 \rightarrow p_1 - p_2 > 0$$

RR: $Z > 1.645$

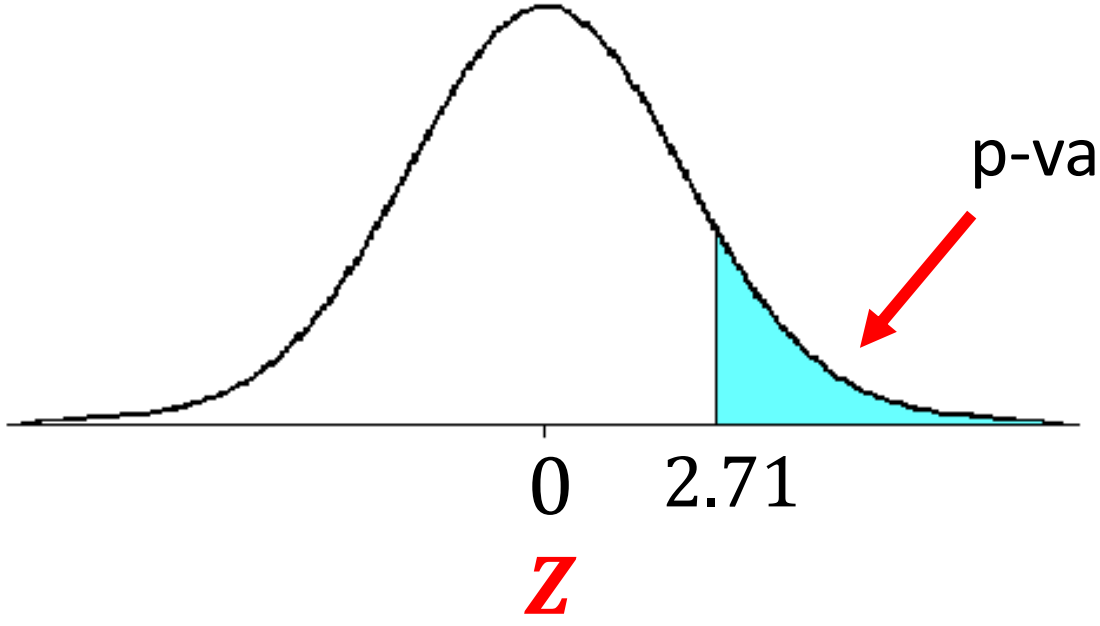


$$Z = \frac{(0.42 - 0.24) - 0}{\sqrt{0.33(0.67)}\sqrt{\frac{1}{100} + \frac{1}{100}}} \approx 2.71$$

reject H_0

We have significant evidence that the drug yields a higher proportion of patients who report good results.

p-value $< \alpha$



p-value interpretation:

If H_0 is true and the drug is truly no more effective than a placebo, then we would only observe this big of a difference in sample proportions (or an even bigger difference) between a treatment group and a placebo group in about 1 in every 294 studies (on average) with sample sizes of 100 each.